1. Clinical Question & Core Hypothesis

Clinical Question

Stromal TIL density correlates with an active immune milieu yet not all TIL-high TNBC achieve pCR after NACT — why does morphological infiltration sometimes fail to translate into effective chemo-cooperative anti-tumor immunity? 2) Tier 1 path used: the single path linking stromal TIL density → IFN-γ–related transcriptional programs → pCR; I built on this because IFNG is the canonical driver of cytotoxic/chemokine programs and HLA-DRA represents tumor/antigen-presentation competence required for effector function. 3) Tier 2 supporting evidence (used to justify mechanisms but not in the core claim): downstream effectors and modulators such as CXCL9/CXCL10 (T cell recruitment), GZMB/PRF1 (cytotoxic effectors), STAT1 (IFN signaling mediator) and IDO1 (immunosuppressive counter-regulator) provide a mechanistic rationale for how IFNG/HLA-DRA status would modulate functional immunity and response to chemo. 4) Endpoint choice: therapy_response (pCR) is the most relevant, measurable neoadjuvant outcome in TNBC and directly tests the proposed immune–chemo cooperation mechanism. 5) What I excluded and why: I limited the core claim to exactly one imaging feature (TIL density) plus two anchor genes (IFNG, HLA-DRA) to keep the hypothesis sharply mechanistic; I excluded TLS as a core variable to avoid conflating two morphological features and relegated additional immune genes (CXCL9, CXCL10, GZMB, PRF1, STAT1, IDO1) to secondary_modifiers for follow-up interrogation?

Working Hypothesis

Are TIL-high TNBCs that fail to achieve pCR after neoadjuvant chemotherapy characterized by deficient IFN-γ signaling and/or impaired HLA-DRA–mediated antigen presentation, rendering infiltrating T cells functionally inert? If so, integrating stromal TIL density with IFNG and HLA-DRA expression (and related IFN-signaling modifiers such as CXCL9, CXCL10, GZMB, PRF1, STAT1, and IDO1) will improve pCR prediction beyond morphology alone.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	IFNG, HLA-DRA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

IFNG: IFNG, HLA-DRA) to keep the hypothesis sharply mechanistic; I excluded TLS as a core variable to avoid conflating two morphological features and relegated additional immune genes (CXCL9, CXCL10, GZMB, PRF1, STAT1, IDO1) to secondary_modifiers for follow-up interrogation.

HLA-DRA: HLA-DRA) to keep the hypothesis sharply mechanistic; I excluded TLS as a core variable to avoid conflating two morphological features and relegated additional immune genes (CXCL9, CXCL10, GZMB, PRF1, STAT1, IDO1) to secondary_modifiers for follow-up interrogation.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	CXCL9	Supporting modifier from knowledge graph
Secondary	CXCL10	Supporting modifier from knowledge graph
Secondary	GZMB	Supporting modifier from knowledge graph
Secondary	PRF1	Supporting modifier from knowledge graph
Secondary	STAT1	Supporting modifier from knowledge graph
Secondary	IDO1	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: retrospective analysis of TNBC cohorts with pCR data using RNA/protein measures of IFNG, HLA-DRA, and secondary modifiers plus TIL density;

Prediction 2: development of a composite predictive score with cross-validated performance metrics (AUC/PR);

Prediction 3: orthogonal validation via multiplex IHC and spatial transcriptomics to link functional signaling with spatial immune context;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- CXCL9: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CXCL10: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.